

Lecture Notes:
Math & Statistics for Data Science
Unit 2 – Linear Algebra

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1 Motivation

- Solving systems of linear equations.
- Describing the geometry of space.
- Foundation for machine learning and deep learning algorithms.

2 Systems of Equations

Learning Goals

- Solve linear systems using algebraic and graphical methods.
- Recognize conditions for uniqueness of solutions.

Content

- **Solving methods:** substitution, elimination, Gaussian elimination.
- **Graphical interpretation:**
 - 2 equations $\rightarrow$ intersecting lines.
 - 3 equations $\rightarrow$ intersecting planes.
- **Conditions for solutions:**

- No solution: parallel lines/planes.
- One solution: unique intersection point.
- Infinite solutions: overlapping.

- **Matrix/vector representation:**

$$\sum_j A_{ij}x_j = b_i$$

Exercises

- Solve 2×2 and 3×3 systems both algebraically and graphically.

3 Vectors as Their Own Entities

Learning Goals

- Understand vectors as distinct from points.
- Perform vector arithmetic and interpret linear combinations.

Content

- **Points vs. vectors:** location vs. magnitude + direction.
- **Notation:** arrows, bold symbols, components.
- **Norm:** $|x| = \sqrt{\sum x_i^2}$.
- **Unit vectors:** normalized vectors, direction only.
- **Vector operations:** addition, scalar multiplication, normalization.
- **Linear combinations and span:**

$$\text{span}(v, w) = \{av + bw \mid a, b \in \mathbb{R}\}.$$

- **Basis vectors:** usually orthonormal; change of basis introduces matrices.

Exercises

- Compute norms, unit vectors, and vector spans in 2D/3D.
- Express vectors in different bases.

4 Matrices

Learning Goals

- Define and manipulate matrices.
- Connect matrices to linear transformations and systems of equations.

Content

- **Notation:** capital letters, rows/columns as vectors.
- **Operations:** addition, scalar multiplication, transpose.
- **Matrix multiplication:**
 - Not commutative.
 - Interpret as transforming vectors.
- **Vector products:**
 - Dot product: $x \cdot y = \sum_i x_i y_i = |x||y| \cos \theta$.
 - Projection and orthogonality.
 - Outer product: row-column multiplication.

Exercises

- Multiply vectors and matrices in Python.
- Demonstrate projections using dot products.

5 More Vectors and Matrices

Content

- Review of basics.
- Equations of lines and planes: $(x - p) \cdot n = 0$.
- **Special matrices:** identity, diagonal, symmetric, unitary.
- **Inverses:**
 - $A^{-1}A = I$.
 - Formula for 2×2 case.
 - Determinant: geometric interpretation (area, volume).
 - Singular matrices ($\det(A) = 0$) have no inverse.
- **Solving with inverses:** $x = A^{-1}b$.

Exercises

- Compute determinants of 2×2 and 3×3 matrices.
- Identify when matrices are singular.
- Solve systems using inverses.

6 Applications

Content

- **Linear regression:**
 - Model: $y = a + bx$.
 - Loss functions (SSR).
 - Solve using systems of equations (normal equations).
- **Linear dependence in data:**
 - Adding variables may not add information.
 - Approximate linear dependence in regression models.

Exercises

- Fit simple regression models with matrix methods.
- Explore collinearity in datasets.

7 Eigenvalues and Eigenvectors

Learning Goals

- Understand eigenvalues and eigenvectors as fundamental invariants of matrices.
- Apply them in practical contexts (physics, ML, statistics).

Content

- **Definition:** $Av = \lambda v$.
- Eigenvector: direction preserved; Eigenvalue: scaling factor.
- **Computation:**
$$\det(A - \lambda I) = 0$$
leads to an n th-degree polynomial.
- **Properties:**
 - Eigenvalues may be real, zero, negative, or complex.
 - Symmetric matrices: real, orthogonal eigenvectors.
 - Diagonalization: $A = VDV^{-1}$.
- **Applications:** matrix powers, PCA.

Exercises

- Find eigenvalues/eigenvectors of 2×2 matrices.
- Diagonalize simple symmetric matrices.
- Use Python to visualize matrix action on vectors.

Pedagogical Notes

- Emphasize geometric intuition: span, orthogonality, eigen-directions.
- Use Python demos for Gaussian elimination, matrix multiplications, regression, and eigenvalue computations.
- Link to data science applications: regression, PCA, dimensionality reduction.